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COMPARATIVE EFFICACY OF CLEARING-DOSE AND SINGLE HIGH-DOSE IVERMECTIN AND DIETHYLCARBAMAZINE AGAINST *WUCHERERIA BANCROFTI* MICROFILAREMIA

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Abstract. To compare the efficacy and tolerability of various combinations of low- and high-dose ivermectin and diethylcarbamazine (DEC), 59 persons with *Wuchereria bancrofti* microfilaremia were enrolled in a double-blinded six-arm clinical trial in Leogane, Haiti. On day 1, study participants were treated with low clearing doses of ivermectin, DEC, or placebo; on day 5 they received 200–400 μg/kg of ivermectin or 6 mg/kg of DEC. Adverse reactions, which were generally mild, occurred more frequently with ivermectin than with DEC. One year after treatment, the geometric mean microfilarial density returned to 0.9% of pretreatment levels for persons who received a total of 420 μg/kg of ivermectin. This rate was significantly lower than 5.6% for persons who were treated with 220 μg/kg of ivermectin ($P = 0.02$) and 9.3% for those receiving 6 or 7 mg/kg of DEC ($P = 0.006$). Persons treated with a clearing dose of ivermectin followed by 6 mg/kg of DEC also had low microfilarial densities (1.7% of pretreatment levels), suggesting an additive or synergistic effect of the two drugs. The addition of a clearing dose neither reduced the severity of adverse reactions nor improved the efficacy of high-dose ivermectin. Community-based intervention trials are now warranted to determine the feasibility and effectiveness of mass chemotherapy with single high-dose ivermectin for the prevention and control of lymphatic filariasis.

Wuchereria bancrofti, the most common cause of human lymphatic filariasis worldwide, infects an estimated 82 million people and causes clinical symptoms of elephantiasis, hydrocele, and adenolymphangitis.[1] Efforts to prevent and control this disease have been limited by the lack of a safe and effective drug that can be given in a single dose. When given in total doses of ≥ 72 mg/kg, diethylcarbamazine (DEC) is frequently effective against the adult worm.[2] However, multiple doses of DEC are usually required because single doses of > 6 mg/kg are poorly tolerated, particularly in persons with high levels of microfilaremia.[2, 3]

Ivermectin, a macrolide antibiotic, has been shown to dramatically reduce microfilaremia levels when given in a single dose and has been considered potentially useful for community-based filariasis control programs.[4, 5] However, the long-term efficacy of a single dose of ivermectin is generally less than that of a full course of DEC; in patients treated with single 20–200 μg/kg doses of ivermectin, microfilaria counts have been observed to return to 13–35% of pretreatment levels within six months, indicating little, if any, impact on the adult worm.[4–6]

A recent study suggested that ivermectin may be effective against the adult worm when given in total doses > 200 μg/kg.[7] To minimize the severity of adverse reactions, Richards and others gave Haitian patients with *W. bancrofti* microfilaremia a 20 μg/kg dose of ivermectin to clear microfilariae five days before a high dose of 200 or 400 μg/kg (200 μg/kg/day on two consecutive days).[7] Surprisingly, the efficacy of ivermectin given in this manner was comparable to that of a standard 12-day course of DEC in clearing microfilariae for up to two years; 400 μg/kg was more effective than 200 μg/kg, although the difference was not statistically significant.[7, 8] It was unclear whether the apparent activity of ivermectin against the adult worm was caused by the higher doses of ivermectin, an added benefit of the low clearing dose, or factors related to the biology of the parasite or the immune response of the human host.

The objectives of the current study were to determine whether the addition of a clearing dose increases long-term efficacy of single high-dose ivermectin, to compare the efficacy and tolerability of ivermectin with that of single-dose DEC, and to assess the efficacy of sequential ivermectin-DEC therapy for treatment of *W. bancrofti* microfilaremia.

SUBJECTS AND METHODS

The study was conducted between October 1990 and October 1991 in the coastal community of Leogane, Haiti, where the prevalence of microfilaremia is approximately 30–35%.[9] After the nature of the study had been explained in Creole and verbal consent had been obtained, 65 microfilaremic men and women, ages 14–65 years, were enrolled. The participants were in good health, had received no anthelminthic therapy during the preceding six months, and were free of malaria as determined by 1–2 Giemsa-stained thick blood smears during the week before the study began. Pregnant and lactating women were excluded.

Study design

To ensure that pretreatment microfilaria levels would be comparable among treatment groups, study participants were stratified by the number of microfilariae per milliliter of venous blood into the following four groups: < 100 (n = 5), 100–499 (n = 26), 500–999 (n = 12), or ≥ 1,000 (n = 22). On day 1, the participants received either placebo or a low clearing dose of ivermectin (20 μg/kg) or DEC (1 mg/kg); on day 5, they received either DEC (6 mg/kg) or high-dose ivermectin (200–400 μg/kg). Within each stratum, the study participants were randomly assigned to one of six treatment groups, with 9–12 persons in each group: group 1 = ivermectin (20 μg/kg) on day 1 followed by a 400-μg/kg dose on day 5; group 2 = ivermectin (20 μg/kg) on day 1 followed by a 200-μg/kg dose on day 5; group 3 = ivermectin (20 μg/kg) on day 1 followed by a 6-mg/kg dose of DEC on day 5; group 4 = DEC (1 mg/kg) on day 1 followed by a 6-mg/kg dose of DEC on day 5; group 5 = placebo on day 1 followed by a 220-μg/kg dose of ivermectin on day 5; group 6 = placebo on day 1 followed by a 6-mg/kg dose of DEC on day 5. The investigators involved in the clinical and parasitologic evaluation were unaware of the treatment group assignments. All medications were taken under observation in the outpatient clinic of Sainte Croix Hospital in Leogane. The study was approved by the institutional review board of the Centers for Disease Control.

Clinical evaluation

Physical examinations were conducted by one of two physicians in the clinic before treatment on day 1 and after treatment on days 2, 3, 5, 6, 7, 10, 120, 180, and 360. The examinations included a medical history, oral temperature, respiration rate, sitting blood pressure, auscultation of the chest, and evaluation of the skin, extremities, and lymph nodes. Male genitalia were examined for abnormalities of the testes, epididymis, and spermatic cord. Persons with significant signs or symptoms after treatment were also evaluated in their homes on days 4, 8, and 9. To measure symptom intensity, a three-point severity scale was developed; the scale included all signs and symptoms that were noted more frequently after treatment than during the pretreatment evaluation. Each sign or symptom (including fever, chills, sweats, headache, lethargy, weakness, myalgias, inguinal tenderness, and genital tenderness) was graded as 0 if not present; 1 if the sign or symptom was noticed by the patient but did not interfere with daily activities; 2 if there was some interference with daily activities; and 3 if the symptoms prevented usual daily activities. Fever was graded as 0 (< 37.8°C), 1 (37.8–38.5°C), 2 (38.6–39.5°C), or 3 (> 39.5°C). A total peak intensity (TPI) score, the sum of the maximum values for the nine signs and symptoms, was calculated for each subject on day 1 (pretreatment) and on post-treatment days 1–4 (postclearing dose) and 5–10 (post-high dose). Adverse reactions, described below, were treated with acetaminophen and, in a few cases, with diphenhydramine. The six treatment groups did not differ significantly in mean age (32 years), sex (58% female), or mean pretreatment temperature (37.4°C).

Laboratory evaluation

Venous blood specimens were obtained between 7:30 PM and 10:00 PM local time before treatment on day 1 and on days 5 (before high-dose medication), 10, 120, 180, and 360. Following the protocol developed for these studies,

TABLE 1

*Pretreatment characteristics and microfilarial (mf) densities of persons participating in a six-arm clinical trial of ivermectin (IV) and diethylcarbamazine (DEC) in Leogane, Haiti, by treatment group**

Group	Treatment Day 1	Treatment Day 5	n	Mean age, years (range)	No. of men (no. with hydroceles)	Geometric mean mf density, mf/ml (95% CI)
1	IV 20 μg/kg	IV 400 μg/kg	9	31.6 (14–52)	3 (0)	805 (386–1,680)
2	IV 20 μg/kg	IV 200 μg/kg	9	30.4 (19–64)	2 (1)	727 (312–1,692)
3	IV 20 μg/kg	DEC 6 mg/kg	11	33.6 (14–65)	4 (0)	855 (390–1,873)
4	DEC 1 mg/kg	DEC 6 mg/kg	9	27.3 (17–54)	4 (2)	696 (293–1,656)
5	Placebo	IV 220 μg/kg	10	35.6 (14–58)	7 (4)	718 (323–1,596)
6	Placebo	DEC 6 mg/kg	11	31.6 (17–63)	5 (3)	571 (284–1,148)
All groups			59	31.8 (14–65)	25 (10)	721 (530–983)

* CI = confidence interval.

microfilaria counts were based on a 0.25-ml sample of venous blood. Blood samples were mixed with 2.5 ml of 10% Teepol (Sigma, St. Louis, MO) and filtered through a 5-μm Nuclepore (Pleasanton, CA) filter. The filters were placed on slides, stained, and examined microscopically as previously described.[7] The number of microfilariae per milliliter of blood was estimated by multiplying the number of observed microfilariae by four.

Serum specimens obtained on days 1 and 10 were tested for glutamic oxaloacetic transaminase (SGOT), glutamic pyruvic transaminase (SGPT), alkaline phosphatase, and serum creatinine. Complete blood counts and urinalyses were performed on days 1 and 10.

The six treatment groups did not differ significantly by pretreatment mean microfilaria count (1,481 organisms/ml), geometric mean microfilarial density (721 organisms/ml), mean hematocrit (38.9%), leukocyte concentration (5,252/mm^3), percent eosinophils (5.4%), eosinophil concentration (282/mm^3), SGOT (40.7 U/ml), SGPT (14.6 U/ml), alkaline phosphatase (45.1 U/ml), or serum creatinine (1.7 mg/dl). Of the 25 men, 10 (40%) had hydroceles (Table 1).

Statistical methods

The chi-square and Fisher's exact tests were used to compare dichotomous variables; continuous variables were compared using the nonparametric Kruskal-Wallis or Wilcoxon rank sum tests. Percentage return of the geometric mean microfilaria density to pretreatment levels was obtained by calculating the post-treatment to pretreatment microfilaria count ratio, multiplying by 100, adding one, taking the log of this value for each individual, calculating the mean for each treatment group, exponentiating this value, and subtracting one.

RESULTS

Study participants had pretreatment microfilaria counts of 36–8,840/ml of venous blood. Five participants (one in each of five treatment groups) had counts less than 100/ml; all but one (80%) were amicrofilaremic one year after treatment. An additional person refused phlebotomy after day 10. These six individuals were removed from further analysis to facilitate eventual comparison with results of similar studies sponsored by the Special Programme in Research and Training in Tropical Diseases (TDR) of the United Nations Development Programme/World Bank/World Health Organization, which excluded subjects with pretreatment microfilaremias of <100/ml. Removal of these persons from the analysis did not substantially change our results or alter our conclusions about the relative efficacy of different treatment regimens.

Tolerance

The adverse reactions most commonly reported during the first 10 days of the study were headache (79.7%), myalgias (78.0%), weakness (52.5%), and lethargy (50.8%); 44.1% of subjects had recorded temperatures of ≥ 37.8°C. Signs and symptoms following treatment lasted from a few hours to five days (median 2 days) and were qualitatively similar for persons receiving ivermectin and DEC.

In the four days following low clearing-dose medication, TPI scores ranged from 0 to 22, with

TABLE 2

Mean total peak intensity scores in a six-arm clinical trial of ivermectin (IV) and diethylcarbamazine (DEC) in Leogane, Haiti, by treatment group

Group	Treatment		Mean total peak intensity score		
	Day 1	Day 5	Pretreatment	Days 1–4	Days 5–10
1	IV 20 μg/kg	IV 400 μg/kg	0.8	8.1	1.0
2	IV 20 μg/kg	IV 200 μg/kg	0.7	10.3	1.3
3	IV 20 μg/kg	DEC 6 mg/kg	0.2	8.5	4.3
4	DEC 1 mg/kg	DEC 6 mg/kg	0.8	1.7	0.8
5	Placebo	IV 220 μg/kg	1.2	2.1	8.2
6	Placebo	DEC 6 mg/kg	1.0	2.5	4.6

a median score of 6. Persons who received a clearing dose of 20 μg/kg of ivermectin had significantly higher TPI scores ($P = 0.003$) and maximum recorded temperatures ($P = 0.003$) than did persons who received a 1-mg/kg clearing dose of DEC (Table 2). As measured by TPI scores, symptoms were significantly more intense following low-dose ivermectin or DEC on day 1 than after higher doses of these drugs on day 5 ($P = 0.001$). Overall TPI scores increased directly with microfilaremia and were highest for patients with pretreatment microfilaria counts > 1,000/ml ($P = 0.02$).

Moderate or severe tenderness of the spermatic cord or testicle, with or without new onset of inguinal adenopathy or tenderness, was observed after treatment in four men, two of whom had hydroceles. Onset of genital tenderness occurred within 96 hr after a 6-mg/kg dose of DEC in three men and within 48 hr after a 20-μg/kg dose of ivermectin in a fourth man.

No abnormal elevations of SGOT, SGPT, or alkaline phosphatase were noted in any participant, and none had significant changes in total leukocyte count or hematocrit. Moderate eosinophilia was observed on day 10 for all groups, ranging from a mean of 319 eosinophils/mm^3 for group 4 to 590 eosinophils/mm^3 for group 5; this difference was not statistically significant.

Efficacy

On day 10, persons receiving only DEC (groups 4 and 6) had significantly higher geometric mean microfilaria counts (20.9 and 55.6/ml, respectively) than did persons receiving ivermectin (0.1–0.2/ml; $P = 0.001$). Similarly, 95.0% of persons who had received only DEC were microfilaria positive on day 10, compared with 12.8% of persons receiving ivermectin ($P < 0.001$) (Figure 1). Study participants who received only DEC tended to have higher microfilaria counts throughout the year of followup. However, by four months post-treatment the percentage of persons who were microfilaria positive was similar for the groups receiving only DEC and ivermectin (80.0% and 82.1%, respectively).

Four months after treatment, an apparent increase in microfilaremia was observed in all treatment groups (Figure 2). Diurnal variation may have been responsible for this finding because only the four-month evaluation was conducted during standard rather than daylight savings time; blood was therefore collected, on average, 1 hr later and thus closer to the period of maximum microfilarial density in Haiti (9:00 PM–3:00 AM).[10]

Twelve months after treatment, the percentage of persons who were microfilaria positive was significantly lower for groups that received 420 μg/kg of ivermectin (group 1) or ivermectin followed by DEC (group 3) than for the other treatment groups ($P = 0.01$ and $P = 0.03$, respectively, by Fisher's exact test) (Figure 1). The mean, geometric mean, and percentage return of geometric mean microfilarial density to pretreatment levels (hereafter referred to as microfilarial density) were highest among persons who received only DEC (groups 4 and 6), lower among those who were treated with 220 μg/kg of ivermectin (groups 2 and 5), and lowest among persons who received either 420 μg/kg of ivermectin (group 1) or 20 μg/kg of ivermectin followed by a single 6-mg/kg dose of DEC (group 3) (Table 3 and Figure 2). Compared with persons in the groups receiving only DEC, those in groups 1 and 3 had significantly lower microfilarial densities ($P = 0.006$ and $P = 0.02$, respectively). In contrast, neither

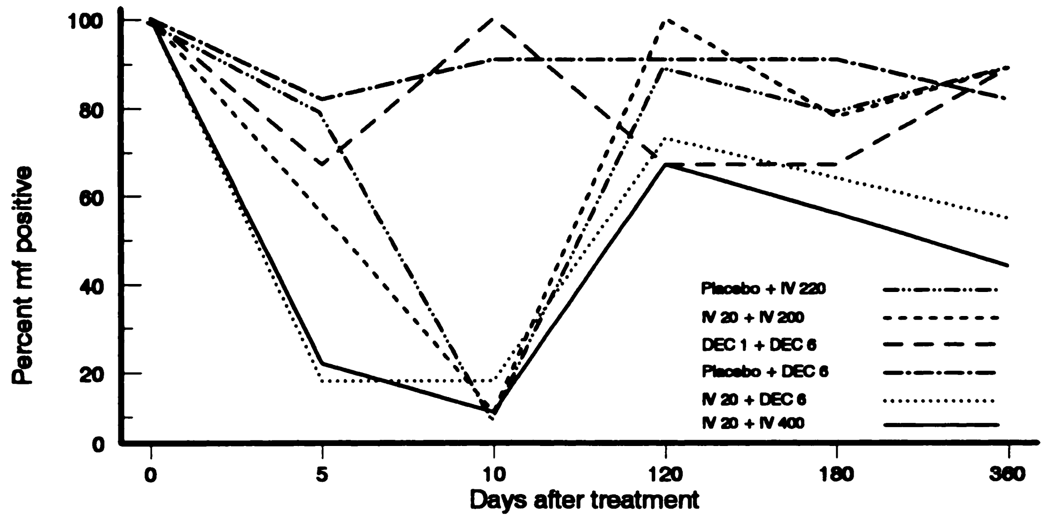

FIGURE 1. Percentage of persons with microfilariae (mf) detected in 0.25 ml of venous blood, by treatment group. IV = ivermectin; DEC = diethylcarbamazine. For a definition of the treatment groups, see Subjects and Methods.

of the groups that received a total of 220 μg/kg of ivermectin had significantly lower microfilarial densities than the groups receiving only DEC (P = 0.8 and P = 0.5 for groups 2 and 5, respectively).

Reduction of microfilaremia one year after treatment was similar for persons who received a single 220-μg/kg dose of ivermectin (group 5) and those who were treated with a 20-μg/kg clearing dose followed five days later by 200 μg/kg of ivermectin (group 2) (Table 3). Compared with persons in these groups, those who received a total of 420 μg/kg of ivermectin (group 1) or 20 μg/kg of ivermectin followed by 6 mg/kg of DEC (group 3) had considerably lower microfilarial densities (P = 0.02 and P = 0.08, respectively).

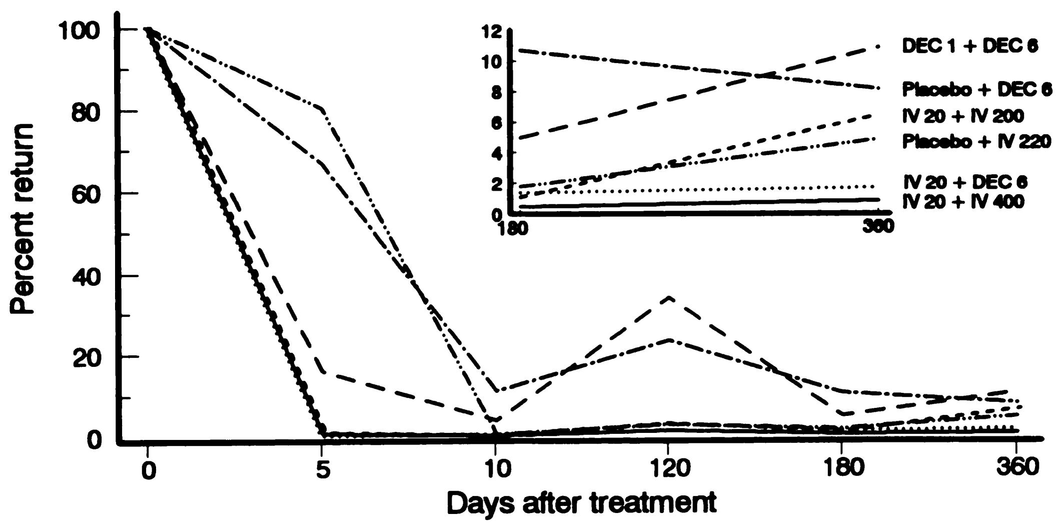

FIGURE 2. Percentage return of geometric mean microfilarial densities to pretreatment levels, by treatment group. Inset shows enlargement of day 180 to day 360. DEC = diethylcarbamazine; IV = ivermectin. For a definition of the treatment groups, see Subjects and Methods.

TABLE 3

*Percentage of persons with microfilaria (mf) detected in venous blood and mean microfilaria counts 12 months after receiving ivermectin or diethylcarbamazine in a six-arm clinical trial in Leogane, Haiti, by treatment group**

Group	n	No. (%) mf-positive	Mean no. of mf/ml	Geometric mean no. of mf/ml (95% CI)	Percent return to pretreatment geometric mean mf/ml (95% CI)
1	9	4 (44)	42.7	4.8 (0.3–25.0)	0.9 (0.0–2.6)
2	9	8 (89)	64.0	29.2 (9.3–87.4)	6.5 (1.7–19.8)
3	11	6 (55)	36.7	7.7 (1.4–30.3)	1.7 (0.3–4.7)
4	9	8 (89)	246.2	76.1 (19.0–296.7)	10.9 (3.7–28.8)
5	10	9 (90)	87.2	27.6 (8.5–84.7)	4.9 (1.5–12.7)
6	11	9 (82)	198.9	33.9 (8.0–134.6)	8.2 (2.5–23.3)

* CI = confidence interval. See Table 1 for information on the groups.

DISCUSSION

Dramatic sustained reductions in microfilaremia were observed in all six groups following treatment with ivermectin or DEC, suggesting that the drugs were active against the adult worm. Although suppression of microfilaremia was less prolonged in persons who received only DEC, the effectiveness of a single 6-mg/kg dose of DEC was considerably greater than expected. One year after treatment, persons who received ivermectin had significantly lower microfilarial densities than did persons who received only DEC, and efficacy of ivermectin was greater at higher doses. The most effective regimen included a single 400-μg/kg dose of ivermectin, resulting in microfilarial densities that were 0.9% of pretreatment levels. This outcome compares favorably with a 0.6–1.7% return to pretreatment levels for Haitians treated with 72 mg/kg of DEC (Eberhard ML, unpublished data).[11] Because study participants were randomly assigned to treatment groups and had comparable demographic characteristics and pretreatment microfilaria counts, differences in these factors cannot explain the observed differences between groups in post-treatment microfilaria densities.

Two treatment regimens, 420 μg/kg of ivermectin (group 1) and ivermectin followed by DEC (group 3), were significantly more effective than other regimens in both reducing microfilarial densities and clearing microfilariae from the blood of infected patients for one year after treatment. The use of a relatively small (0.25 ml) sample of venous blood, an insensitive measure for the presence of microfilaria,[12] may have resulted in overestimation of the proportion of subjects who were amicrofilaremic, but this overestimation would have been expected to occur to a similar extent in all six treatment groups.

The association we observed between pretreatment microfilaria counts and the severity of adverse reactions has been well documented.[4, 5, 7, 13] Some[4, 5] but not all[13] studies have also shown that adverse reactions tend to be more severe as the dose of ivermectin increases. In our study population, the severity of adverse reactions did not increase with higher doses of ivermectin; although the number of study participants was small, use of a low clearing dose of ivermectin was not associated with decreased symptom severity. This finding suggests that a clearing dose may not be necessary in community-based control programs in Haiti if ivermectin is distributed in single doses of ≤ 220 μg/kg. Whether a single 400-μg/kg dose of ivermectin given without a preceding clearing dose would be associated with more severe side effects is not yet known.

Although low-dose ivermectin has been considered potentially useful for clearing microfilaria and reducing overall symptom severity,[7] we found that a 1-mg/kg clearing dose of DEC was associated with fewer side effects than a 20-μg/kg dose of ivermectin. Furthermore, following a 6-mg/kg dose of DEC on day 5, TPI scores were lower for persons who had received a clearing dose of DEC (group 4) than for those who had received ivermectin (group 3) on day 1 ($P = 0.06$). These data suggest that in areas where a clearing dose is considered necessary, DEC may be preferred over ivermectin.

The efficacy of ivermectin was not enhanced by the addition of a clearing dose of ivermectin. It is therefore likely that the total dose (≥ 220 μg/kg) of ivermectin, rather than an added effect of the clearing dose, was responsible for the efficacy of ivermectin reported by Richards and others in the same Haitian community.[7] Our findings also confirm initial observations of these investigators that the long-term efficacy of ivermectin is greater at total doses of 420 μg/kg than at 220 μg/kg.

One year after treatment, persons who had received a 20-μg/kg dose of ivermectin followed by 6 mg/kg of DEC had surprisingly low microfilarial densities. Other investigators have suggested that ivermectin and DEC may act in an additive or synergistic fashion against adult *W. bancrofti*,[14] but little is known about possible mechanisms of action or the degree to which this may occur. In Haiti, a clearing dose of ivermectin appears to enhance the suppressive effect of a standard 12-day course of DEC for up to two years after treatment.[8] If the effects of ivermectin and DEC are additive when the drugs are given simultaneously, single-dose therapy with the combined drugs may be desirable for community use. Because the pharmacokinetics and mode of action of the two drugs appear to differ,[15–17] it is possible that combined low-dose ivermectin and DEC would produce fewer side effects than a single 400-μg/kg dose of ivermectin. Additional work is needed to evaluate the safety and efficacy of combined ivermectin-DEC against *W. bancrofti* microfilaremia and to determine whether the drugs interact synergistically against the adult worm.

Our findings strongly suggest that ivermectin, either alone or in combination with DEC, will play an increasingly important role in community-based efforts to prevent and control bancroftian filariasis. The extent to which relatively small but statistically significant differences in efficacy between treatment regimens are meaningful for the control of filariasis remains to be determined. The optimum dose, treatment interval, and method of drug delivery are unlikely to be the same in all *W. bancrofti*–endemic areas and may be influenced by the strain of the parasite, intensity of exposure, the immunologic response of the human host, resources of the local health system, and local perceptions of acute and chronic bancroftian disease. For community-based prevention programs, however, the most successful drug interventions will be those that can be administered in a single oral dose, are associated with few serious side effects, and result in sustained profound suppression of microfilaremia.

Before mass treatment with ivermectin is recommended, the tolerability of a single 400-μg/kg dose of ivermectin given without a clearing dose should be established. If tolerability is comparable to that of a single 200-μg/kg dose of ivermectin, regular (e.g., yearly) administration of 400 μg/kg of ivermectin, or perhaps a combination of ivermectin and DEC, would appear to be the most promising strategy for control of lymphatic filariasis in Haiti. Community-based intervention trials are now warranted to determine the feasibility of this approach, its impact on transmission of *W. bancrofti*, and its long-term effectiveness in preventing the debilitating clinical manifestations of chronic *W. bancrofti* infection.

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REFERENCES

1. WHO Expert Committee on Filariasis, 1984. Fourth report. *World Health Organ Tech Rep Ser 702.*
2. Ottesen EA, 1985. Efficacy of diethylcarbamazine in eradicating infection with lymphatic-dwelling filariae in humans. *Rev Infect Dis 7*: 341–356.
3. Sundaram RM, Koteswara Rao N, Krishna Rao C, Krishna Rao P, Rao CK, 1974. Studies on bancroftian filariasis control with diethylcarbamazine: I. Frequency and nature of drug reactions. *J Commun Dis 6*: 290–300.
4. Kumaraswami V, Ottesen EA, Vijayasekaran V, Uma Devi S, Swaminathan M, Aziz MA, Sarma GR, Prabhakar R, Tripathy SP, 1988. Ivermectin for the treatment of Wuchereria bancrofti filariasis: efficacy and adverse reactions. *JAMA 259:* 3150–3153.
5. Ottesen EA, Vijayasekaran V, Kumaraswami V, Perumal Pillai SV, Sadanandam A, Frederick S, Prabhakar R, Tripathy SP, 1990. A controlled trial of ivermectin and diethylcarbamazine in lymphatic filariasis. *N Engl J Med 322:* 1113–1117.
6. Diallo S, Aziz MA, Ndir O, Badiane S, Bah IB, Gaye O, 1987. Dose-ranging study of ivermectin in treatment of filariasis due to Wuchereria bancrofti. *Lancet i*: 1030.
7. Richards FO, Eberhard ML, Bryan RT, McNeeley DF, Lammie PJ, McNeeley MB, Bernard Y, Hightower AW, Spencer HC, 1991. Comparison of high-dose ivermectin and diethylcarbamazine for activity against bancroftian filariasis in Haiti. *Am J Trop Med Hyg 44:* 3–10.
8. Eberhard ML, Hightower AW, McNeeley DF, Lammie PJ, 1992. Long-term suppression of microfilaraemia following ivermectin treatment. *Trans R Soc Trop Med Hyg 86:* 287–288.
9. Raccurt CP, Lowrie RC Jr, Katz SP, Duverseau YT, 1988. Epidemiology of Wuchereria bancrofti in Leogane, Haiti. *Trans R Soc Trop Med Hyg 82:* 721–725.
10. Miller MJ, Ratard RC, McNeeley DF, 1976. Filariasis in Haiti. *J Parasitol 62:* 845–847.
11. Eberhard ML, Dickerson JW, Hightower AW, Lammie PJ, 1991. Bancroftian filariasis: long-term effects of treatment with diethylcarbamazine in a Haitian population. *Am J Trop Med Hyg 45:* 728–733.
12. Eberhard ML, Roberts JM, Lammie PJ, Lowrie RC, 1988. Comparative densities of Wuchereria bancrofti microfilaria in paired samples of capillary and venous blood. *Trop Med Parasitol 39:* 295–298.
13. Roux J, Perolat Ph, Cartel JL, Boutin JP, Sechan Y, Lariviere M, Aziz MA, 1989. Etude de l'ivermectine pour le traitement de la filariose lymphatique due a Wuchereria bancrofti var. pacifica en polynesie francaise. *Bull Soc Pathol Exot Filiales 82:* 72–81.
14. Weil GJ, Lammie PJ, Richards FO, Eberhard ML, 1991. Changes in circulating parasite antigen levels after treatment of bancroftian filariasis with diethylcarbamazine and ivermectin. *J Infect Dis 164:* 814–816.
15. Gustafsson LL, Beermann G, Abdi YA, 1987. *Handbook of Drugs for Tropical Parasitic Infections.* London: Taylor and Francis.
16. Fink DW, Porras AG, 1989. Pharmacokinetics of ivermectin in animals and humans. Campbell WC, ed. *Ivermectin and Abamectin.* New York: Springer-Verlag, 113–130.
17. Turner MJ, Schaeffer JM, 1989. Mode of action of ivermectin. Campbell WC, ed. *Ivermectin and Abamectin.* New York: Springer-Verlag, 73–88.